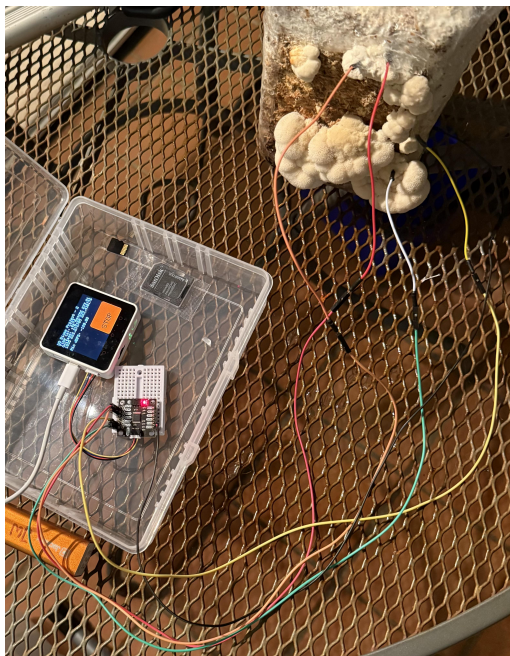
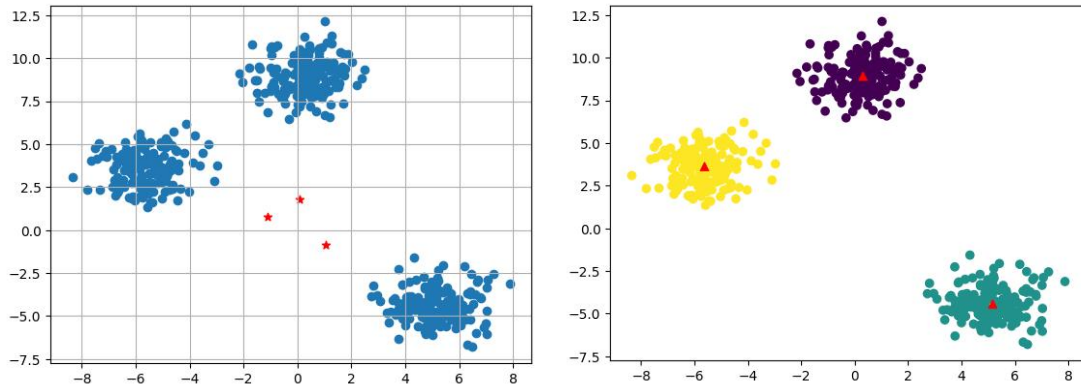
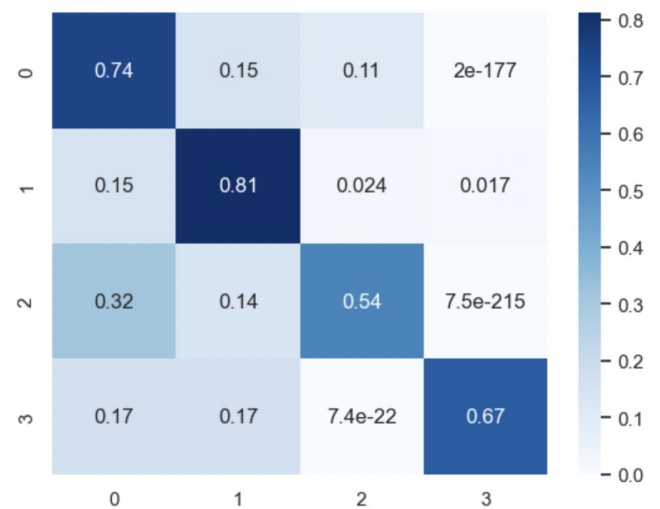


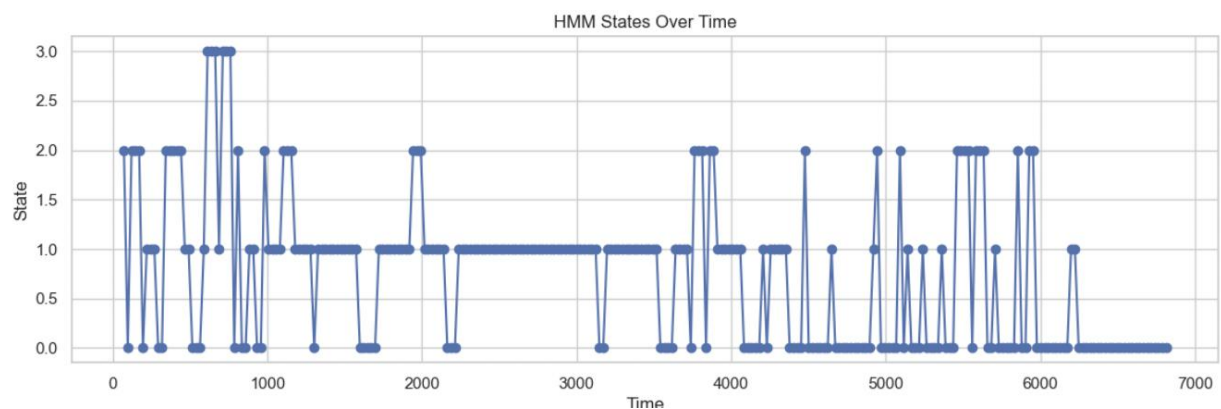
KMeans Clustering



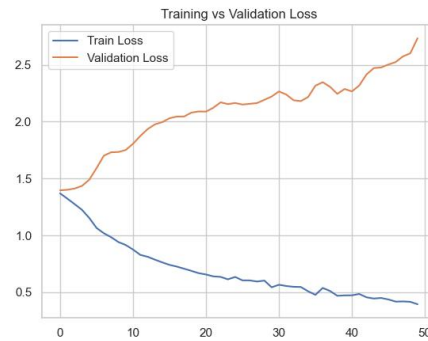
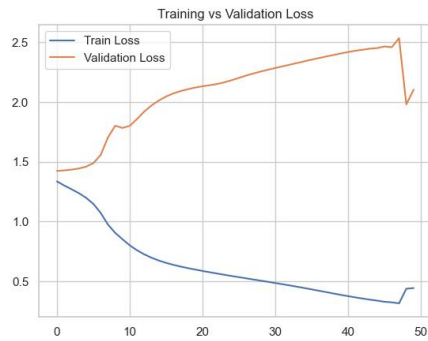
Time Series Data Source



Correlation - Heat Map



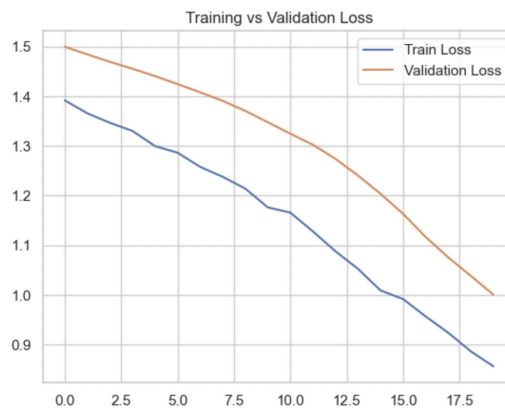
Improvements through adjust some parameters



1. more epoch/data train more overfit
2. Dropout added Better but still overfit The model is becoming increasingly "**memorized**" of the training data, but is getting **worse at handling new data**.

After K-Fold Cross Validation

The model learns more **general rules**, rather than just memorizing the training set.



Evaluation

	fold	accuracy	f2_score	final_train_loss	final_val_loss
0	1	0.045455	0.014205	1.333765	1.412486
1	2	0.931818	0.918378	1.175839	1.147721
2	3	0.500000	0.434783	1.052842	1.199337
3	4	0.386364	0.369009	0.915266	1.360821
4	5	0.659091	0.645913	0.855991	1.000426
Average Accuracy: 0.5045454545454545					
Average F2 Score: 0.4764574776397474					

DECO3801 2026S1 Project 19 - Lucky SHU
Additional Materials for the Final Exhibit's Model Part

Demo for confirming Pattern to prove in models in Lucky's Modeling Notebook.

